

JEE MAIN REVISION MASTER: D AND F BLOCK ELEMENTS

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 236 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	D And F Block Elements — Core Principles & Governing Laws	143 PYQs	Core Principle
Very High	D And F Block Elements — Applied Cases, Reactions & Traps	75 PYQs	Applied Topics
Very High	D And F Block Elements — Standard Calculations & Problem Archetypes	28 PYQs	Quantitative Analysis

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
D And F Block Elements — Core Principles & Governing Laws	$\text{KMnO}_4, \text{Ce}^{+4}, \text{Eu}^{+3}, \text{Tb}^{+3}, \text{Lu}^{+3} \rightarrow \text{MnO}_4^{2-}$
D And F Block Elements — Applied Cases, Reactions & Traps	$[\text{Fe}(\text{CN})_6]^{4-}, \text{MnO}_2, \text{KMnO}_4$
D And F Block Elements — Standard Calculations & Problem Archetypes	$\text{VO}_2^+, \text{MnO}_4^-, [\text{Fe}(\text{CN})_6]^{4-}, \text{Fe}^{2+}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 D And F Block Elements — Core Principles & Governing Laws (Very High Yield — 143 PYQs)

Core Principle: Fundamental definitions, core postulates, and governing physical/chemical relations in D And F Block Elements.

■ **Caution / JEE Pitfall:** Carefully verify units, standard reference states, and sign conventions in D And F Block Elements.

■ **Speed Trick:** Focus on high-yield formulas and standard limiting approximations tested in D And F Block Elements PYQs.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "Given below are two statements : Statement I : The number of pairs among $[Ti^{4+}, V^{2+}], [V^{2+}, Mn^{2+}], [Mn^{2+}, Fe^{3+}]$ and $[V^{2+}], \dots$ " [Ans: A]
- [2026 Evening Shift] Q#1 (MCQ): "Consider the following statements about manganate and permanganate ions. Identify the correct statements. A. The geometry of both manganate ..." [Ans: N/A]
- [2026 Morning Shift] Q#2 (MCQ): "Which of the following is NOT a physical or chemical characteristics of interstitial compounds?..." [Ans: N/A]

3.2 D And F Block Elements — Applied Cases, Reactions & Traps (Very High Yield — 75 PYQs)

Core Principle: Special cases, experimental observations, and conceptual traps designed to test deep understanding of D And F Block Elements.

■ **Caution / JEE Pitfall:** Read questions carefully for keywords like 'incorrect', 'not true', or non-standard conditions.

■ **Speed Trick:** Elimination of extreme or unphysical answer choices improves accuracy in assertion-reason and multiple-statement questions.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "A paper dipped in a dil. H_2SO_4 solution of 'X' upon treatment with SO_2 gas turns into green. The compound 'X' is ..." [Ans: D]
- [2026 Morning Shift] Q#1 (MCQ): "The correct statements among the following are, A. $Mo(VI)$ and $W(VI)$ are less stable than $Cr(VI)$. B. Ce^{4+} and Tb^{4+} are oxida..." [Ans: N/A]
- [2026 Evening Shift] Q#1 (MCQ): "On heating a mixture of common salt and $K_2Cr_2O_7$ in equal amount along with concentrated H_2SO_4 in a test tube, a gas is evolved..." [Ans: C]

3.3 D And F Block Elements — Standard Calculations & Problem Archetypes (Very High Yield — 28 PYQs)

Core Principle: Quantitative numerical problem-solving techniques and standard question patterns frequently seen in D And F Block Elements.

■ **Caution / JEE Pitfall:** Watch for algebraic rounding, stoichiometric balancing, and boundary condition checks.

■ **Speed Trick:** Dimensional consistency checks and ratio comparisons can quickly eliminate incorrect options.

Representative Source PYQs:

- [2026 Morning Shift] Q#1 (Numerical): "Consider the following reactions :
$$\begin{aligned} &NaCl + K_2Cr_2O_7 + H_2SO_4 \rightarrow A + KHSO_4 + NaHSO_4 + H_2O \\ &A + NaOH \rightarrow \dots \end{aligned}$$
" [Ans: 13]
- [2026 Morning Shift] Q#2 (MCQ): "Correct statements from the following are A. Potassium dichromate is an oxidising agent and it oxidises $FeSO_4$ to $Fe_2(SO_4)_3$ i..." [Ans: N/A]
- [2026 Evening Shift] Q#4 (MCQ): "Consider $|x|$ is the difference in oxidation states of Mn in highest manganese fluoride and highest manganese oxide. The ions with $|x|$ nu..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 8th April Evening Shift (Q#1)

Given below are two statements : Statement I : The number of pairs among $[\text{Ti}^{4+}, \text{V}^{2+}]$, $[\text{V}^{2+}, \text{Mn}^{2+}]$, $[\text{Mn}^{2+}, \text{Fe}^{3+}]$ and $[\text{V}^{2+}, \text{Cr}^{2+}]$ in which both ions are coloured is 3 . Statement II : The number of pairs among $[\text{La}^{3+}, \text{Yb}^{2+}]$, $[\text{Lu}^{3+}, \text{Ce}^{4+}]$ and $[\text{Ac}^{3+}, \text{Lr}^{3+}]$ ions in which both are diamagnetic is 3 . In the light of the above statements, choose the correct from the options given below :

- (A) Both Statement I and Statement II are correct **[Correct]**
- (B) Both Statement I and Statement II are incorrect
- (C) Statement I is correct but Statement II is incorrect
- (D) Statement I is incorrect but Statement II is correct

Correct Answer: A

Solution Analysis: Coloured ions have unpaired electron. (i) $\text{T}_{(i)}^{+4} = 4s^0 3d^0$ $n=0$ (ii) $\text{V}^{+2} = 4s^0 3d^3$ $n=3$ (iii) $\text{Mn}^{+2} = 4s^0 3d^5$ $n=5$ (iv) $\text{Fe}^{+3} = 4s^0 3d^5$ $n=5$ (v) $\text{Cr}^{+2} = 4s^0 3d^4$ $n=4$ The absence of unpaired electron makes the element diamagnetic.
$$\begin{aligned} & \text{La}^{+3} = [\text{Xe}] 5d^0 6s^0 \text{ \& } 4f^{+2} = [\text{Xe}] 4f^{14} 6s^0 \text{ \& } \\ & \text{Lu}^{+3} = [\text{Xe}] 4f^{14} 5d^0 6s^0 \text{ \& } \text{Ce}^{+4} = [\text{Xe}] 4f^0 5d^0 6s^0 \text{ \& } \text{Ac}^{+3} = [\text{Rn}] 6d^0 7s^0 \end{aligned}$$

Benchmark Problem 2: JEE Main 2026 (Online) 5th April Evening Shift (Q#1)

A paper dipped in a dil. H_2SO_4 solution of 'X' upon treatment with SO_2 gas turns into green. The compound 'X' is :

- (A) KI-starch
- (B) KMnO_4
- (C) $\text{Pb}(\text{CH}_3\text{COO})_2$
- (D) $\text{K}_2\text{Cr}_2\text{O}_7$ **[Correct]**

Correct Answer: D

Solution Analysis: The correct answer is Option D: $\text{K}_2\text{Cr}_2\text{O}_7$. A paper dipped in dilute H_2SO_4 solution of X turns green when treated with SO_2 gas. This means that X must be a substance which changes to green on reduction by SO_2 . Now, SO_2 is a reducing agent . Acidified potassium dichromate, that is, $\text{K}_2\text{Cr}_2\text{O}_7$ in dilute H_2SO_4 , is orange in colour and gets reduced to Cr^{3+} , which is green . The reaction is:

$$\text{Cr}_2\text{O}_7^{2-} + 3\text{SO}_2 + 2\text{H}^+ \rightarrow 2\text{Cr}^{3+} + 3\text{SO}_4^{2-} + \text{H}_2\text{O}$$
 So, the orange dichromate paper turns green. Now check other options briefly

Benchmark Problem 3: JEE Main 2026 (Online) 21st January Morning Shift (Q#1)

Consider the following reactions : $\text{NaCl} + \text{K}_2\text{Cr}_2\text{O}_7 + \text{H}_2\text{SO}_4 \rightarrow \text{A} + \text{KHSO}_4 + \text{NaHSO}_4 + \text{H}_2\text{O}$ $\text{A} + \text{NaOH} \rightarrow \text{B} + \text{NaCl} + \text{H}_2\text{O}$ $\text{B} + \text{H}_2\text{SO}_4 + \text{H}_2\text{O}_2 \rightarrow \text{C} + \text{Na}_2\text{SO}_4 + \text{H}_2\text{O}$ In the product ' C ' , ' X ' is the number of O_2^{2-} units, ' Y ' is the total number oxygen atoms present and ' Z ' is the oxidation state of Cr . The value of $X+Y+Z$ is _____ .

Correct Answer: 13

Solution Analysis: The first reaction is the well-known Chromyl chloride test . When a chloride salt like sodium chloride (NaCl) is heated with potassium dichromate ($\text{K}_2\text{Cr}_2\text{O}_7$) and concentrated sulfuric acid (H_2SO_4), deep red vapors of chromyl chloride are evolved. $4\text{NaCl} + \text{K}_2\text{Cr}_2\text{O}_7 + 6\text{H}_2\text{SO}_4 \rightarrow 2\text{CrO}_2\text{Cl}_2 + 2\text{KHSO}_4 + 4\text{NaHSO}_4 + 3\text{H}_2\text{O}$ Thus, Product A is Chromyl chloride (CrO_2Cl_2) . When chromyl chloride gas is passed through an aqueous solution of sodium hydroxide (NaOH), a yellow solution of sodium chromate is formed. $\text{CrO}_2\text{Cl}_2 + 4\text{NaOH} \rightarrow$